

Agent-Based Infrastructures for Diachronic Digital Humanities

Cohort

Agents for retrieving, comparing and documenting textual
evidence

Shu-Kai Hsieh¹² · Da-Chen Lian¹² · Chunki Lim¹ · Lang-Ching Yeh¹

¹ Graduate Institute of Linguistics, National Taiwan University, Taipei, Taiwan

² Humanistic-AI lab, College of Liberal Arts, National Taiwan University, Taipei, Taiwan

Acknowledgement

Thank you, Michael Radich

Texts

A modified, selected
CBETA-derived corpus

Catalogue

Translator labels and
mixed historical groups

Vocabulary

Curated strings for
Dharmarakṣa research

Cohort builds the profiles and calculates the rankings.

Why we built Cohort

Let researchers direct agents and inspect the evidence behind their proposals.

Today's presentation

Use Buddhist texts to show the functions, where results can mislead, and what the researcher decides.

We are demonstrating research tools, not reporting a new translator attribution.

Agent support for diachronic research

Study texts and their relationships across time.

Corpus retrieval

Find relevant texts and passages.

Semantic analysis

Find passages with related content.

Alignment

Locate matching wording and positions.

Interpretation

Propose explanations and alternatives.

Provenance

Keep sources and checks attached to proposals.

Researcher decisions

Inspect the evidence; accept or reject proposals.

Functions

Corpus

Find passages and read them in context.

Vocabulary comparison

Compare a text's wording with texts labelled by translator.

Inquiry

Give agents a research question and inspect their actions.

Findings

Read the agents' proposals, citations and checks.

Graph

Follow links between sources, proposals and decisions.

Example research questions

Two Lotus translations

Dharmarakṣa 竺法護 · T0263
Kumārajīva 鳩摩羅什 · T0262

Can we find related passages
when the wording differs?

Text and later commentary

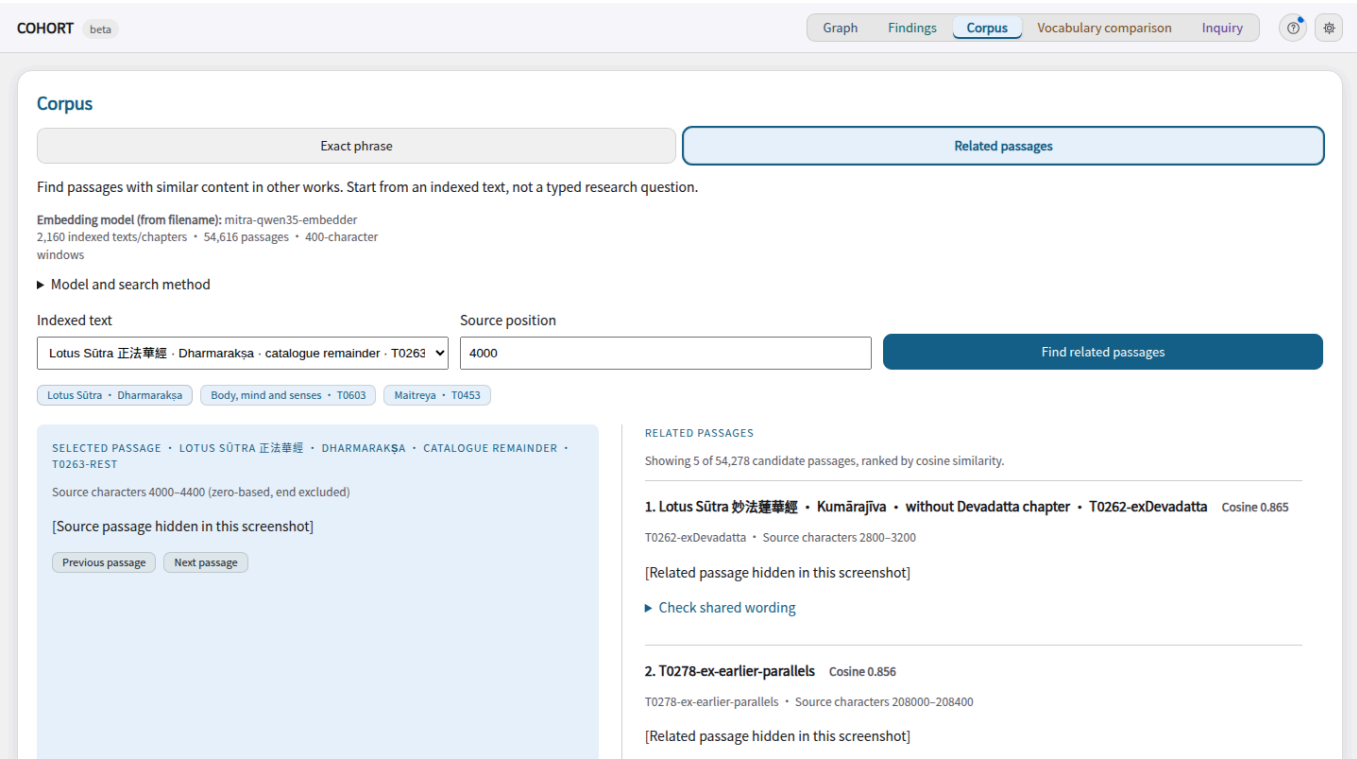
Yin chi ru jing 陰持入經 · T0603
Body, mind and senses

Could shared wording mislead us
about who translated this text?

Known relationships let us inspect what the tools get right and wrong.

Find passages and read them in context.

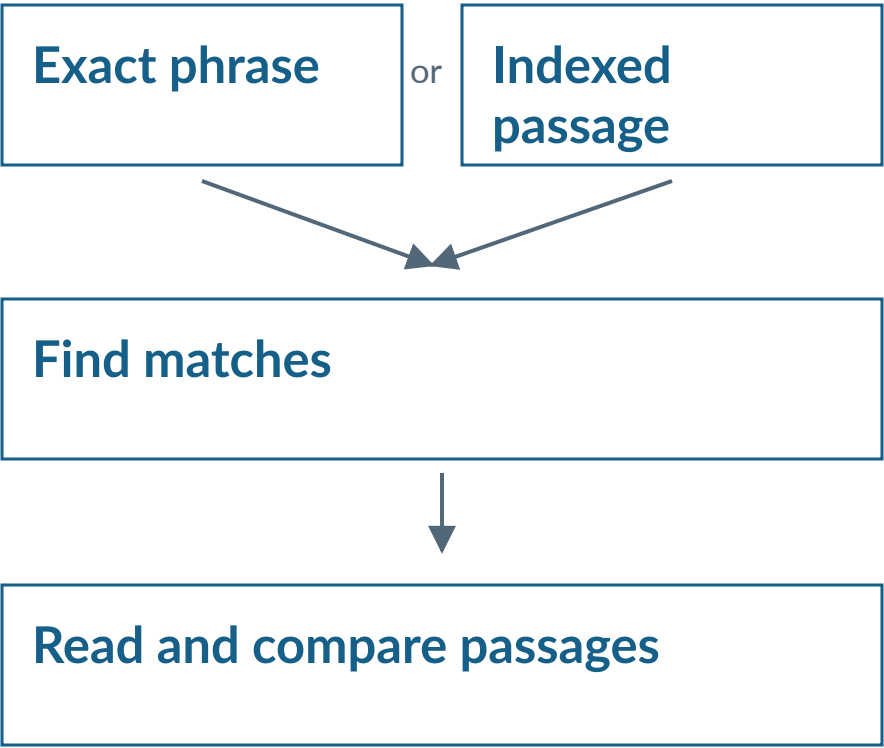
Interface



Exact search or stored-vector retrieval.

Screenshot: source passages hidden where applicable. Diagram: schematic.

How to use it



Browsing does not add graph records.

How related passages are found

mitra-qwen35-embedder

Selected passage

Use its stored vector

Cosine similarity



Compare with other works

Related passages



Rank candidates to read

Corpus → Related passages; agents can call `semantic_neighbors`.

Model name comes from the index filename; its exact revision is unrecorded.

Example: related content, little shared wording

Earlier Lotus translation

Dharmarakṣa · T0263-rest
Source position 4000

cosine 0.865



Later Lotus translation

Kumārajīva · T0262-exDevadatta
Source position 2800

Longest exact shared sequence: 3 Chinese characters

Shared wording is checked separately, with positions in both sources.

What vocabulary comparison calculates

Vocabulary list

Choose strings to count



Target text

Count those strings



Group profiles

Texts pooled by
catalogue label



Compare usage → rank the catalogue groups

13 usable groups in this dataset. The target's whole work is withheld.

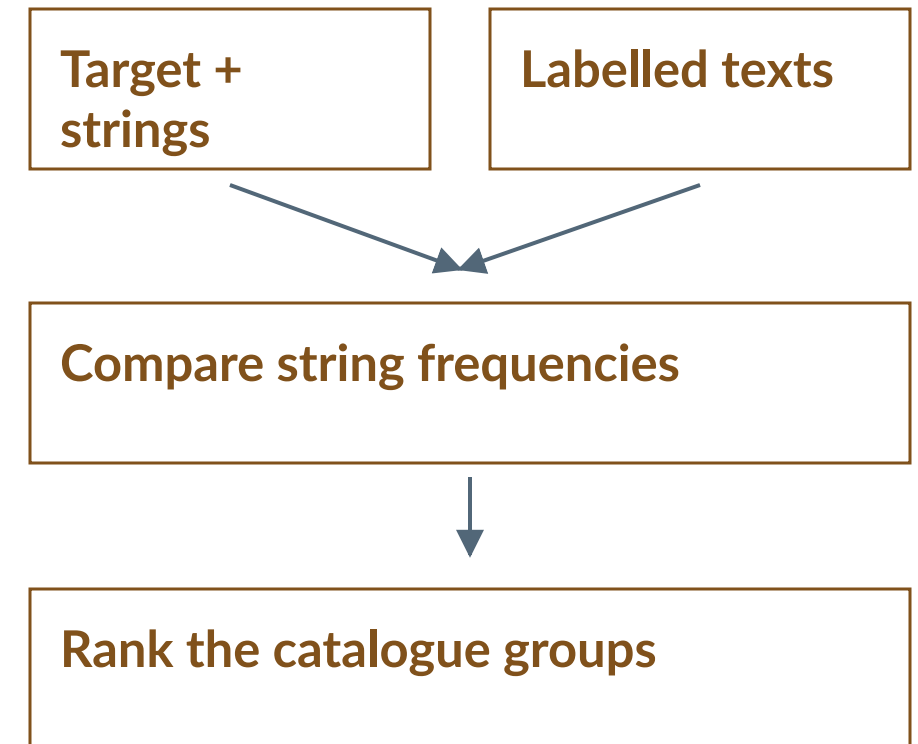
Compare a text's wording with texts labelled by translator.

Interface

Exclude a comparison text and repeat.

Screenshot: source passages hidden where applicable. Diagram: schematic.

How to use it



The vocabulary selects which strings count.

T0603: remove the later commentary

With T1694

First: Other material before
Dharmarakṣa 竺法護

Margin over next group: 1.696



Without T1694

First: An Shigao 安世高

Margin over next group: 0.096

T1694 is the commentary on T0603. Its wording affects the ranking.

This agrees with the accepted attribution after exclusion; it does not establish one.

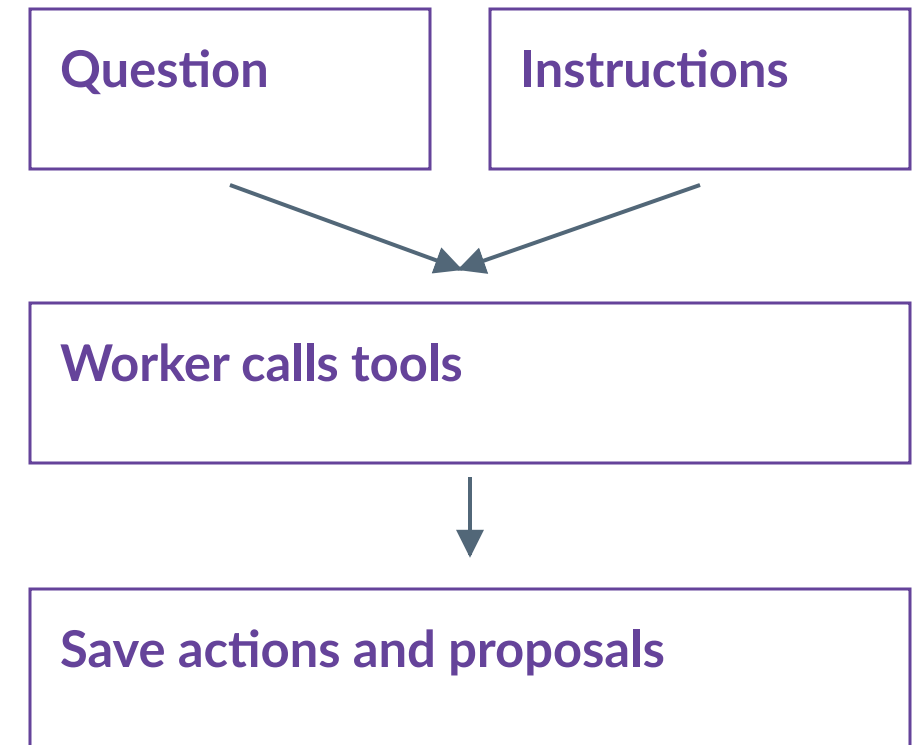
Give agents a research question and inspect their actions.

Interface

The screenshot shows the 'Inquiry' tab in the COHORT beta application. The interface is divided into several sections:

- Header:** COHORT beta, with tabs for Graph, Findings, Corpus, Vocabulary comparison, and Inquiry (selected).
- Inquiry Panel:**
 - 1. Choose or record a research question:** Includes a 'Cancel' button and a prompt to 'Choose an example, then edit and record it.' Below are three buttons: 'Find related passages: T0603', 'T0453: find a parallel', and 'Test recurring formulae'.
 - Research question:** A text box containing 'Which passages are related to the Yin chi ru jing (T0603), and what might explain the relationship?'.
 - Research instructions:** A text box containing 'Find related passages, compare shared wording, and consider quotation, commentary and recurring formulas. Report source positions and limitations.'.
 - Record question:** A large purple button.
 - Saved questions (4):** A section with a 'Show hidden questions (3)' button.
 - 2. Start a new agent run:** A section at the bottom.

How to use it

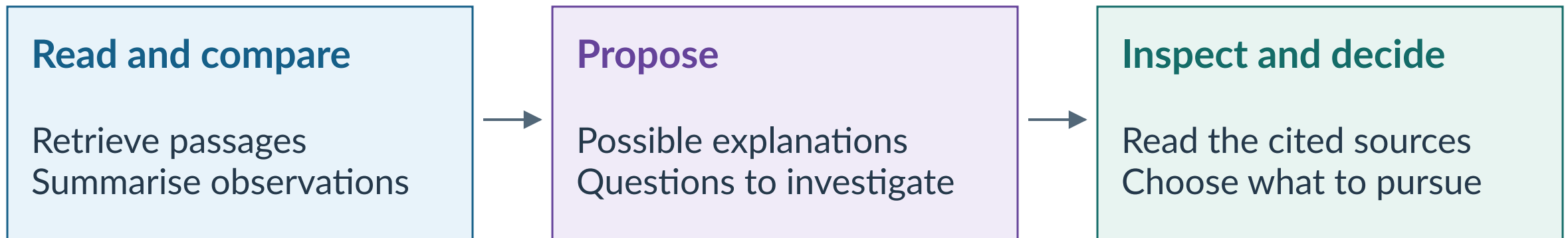


A separate reviewer records checks.

Reopen completed work in Run results.

From passages to research proposals

Agents can compare material, suggest explanations and propose follow-up searches.



Graph

Keeps proposals linked to passages, checks and researcher decisions.

A citation that resolves can still support a weak interpretation.

Read the agents' proposals, citations and checks.

Interface

The screenshot shows the COHORT beta interface. At the top, there are tabs for Graph, Findings (selected), Corpus, Vocabulary comparison, and Inquiry. Below the tabs, it says 'Hypotheses 8' and 'Claims and conjectures, newest first. Not ranked by confidence.' The main area displays a list of hypotheses, each with a status label (CONJECTURE or CLAIM), a description, and a set of tags (ATTENDED, A2 EXACT SPAN MATCHED, 2 ATTESTING, 2 WITNESSES, INDEPENDENT, PROSPECTIVE QUERY NOT YET RUN, DOSSIER).

COHORT beta

Graph **Findings** Corpus Vocabulary comparison Inquiry

Hypotheses 8

Claims and conjectures, newest first. Not ranked by confidence.

CONJECTURE The close relationship between Yin chi ru jing (T0603) and T1694 reflects textual dependence or shared formulaic stock — quotation, commentary, or reuse of the same base exposition — rather than independent composition; whether it also reflects a shared hand cannot be settled from this corpus alone.

ATTENDED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

CLAIM Yin chi ru jing (T0603) and T1694 reproduce the same extended wording, including the eighteen-base exposition and long doctrinal runs up to 279 characters, with 64.45% of T0603's ten-character strings found in T1694.

ATTENDED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CONJECTURE The apparent similarity between T0603 and Other material before Dharmarakṣa 竺法護 (pre-Dhr-other) translator vocabulary is driven by repeated material shared with T1694: once T1694 is withheld from the comparison profiles, the leaning for T0603 collapses to no meaningful distinction between An Shigao 安世高 (ASg) and Other material before Dharmarakṣa 竺法護 (pre-Dhr-other).

ATTENDED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

CLAIM Yin chi ru jing (T0603) reproduces wording also found in T1694 at length, including a 128-character run enumerating the eighteen bases and a 279-character run, with 64.45% of T0603's ten-character strings found in T1694.

ATTENDED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CLAIM T0603 (陰持入經) and T1694 reproduce the same extended wording verbatim, including a 128-character run beginning 四耳五聲六識 and a 279-character run on non-extinction/purity; 64.5% of T0603's ten-character strings occur in T1694.

ATTENDED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT

CONJECTURE The Other material before Dharmarakṣa 竺法護 (pre-Dhr-other) leaning of T0603 (陰持入經) under Radich's vocabulary depends on T1694 sharing its wording: with T1694 withheld the leaning disappears (An Shigao 安世高 (ASg) ahead by margin 0.096, gap 3.4, i.e. no evidence either way), so the corpus as configured cannot settle whether An Shigao translated T0603.

ATTENDED A2 EXACT SPAN MATCHED 2 ATTESTING • 2 WITNESSES • INDEPENDENT PROSPECTIVE QUERY NOT YET RUN DOSSIER

Follow a citation into Graph to inspect its source.

Screenshot: source passages hidden where applicable. Diagram: schematic.

How to use it

Open a saved proposal

Cited passages

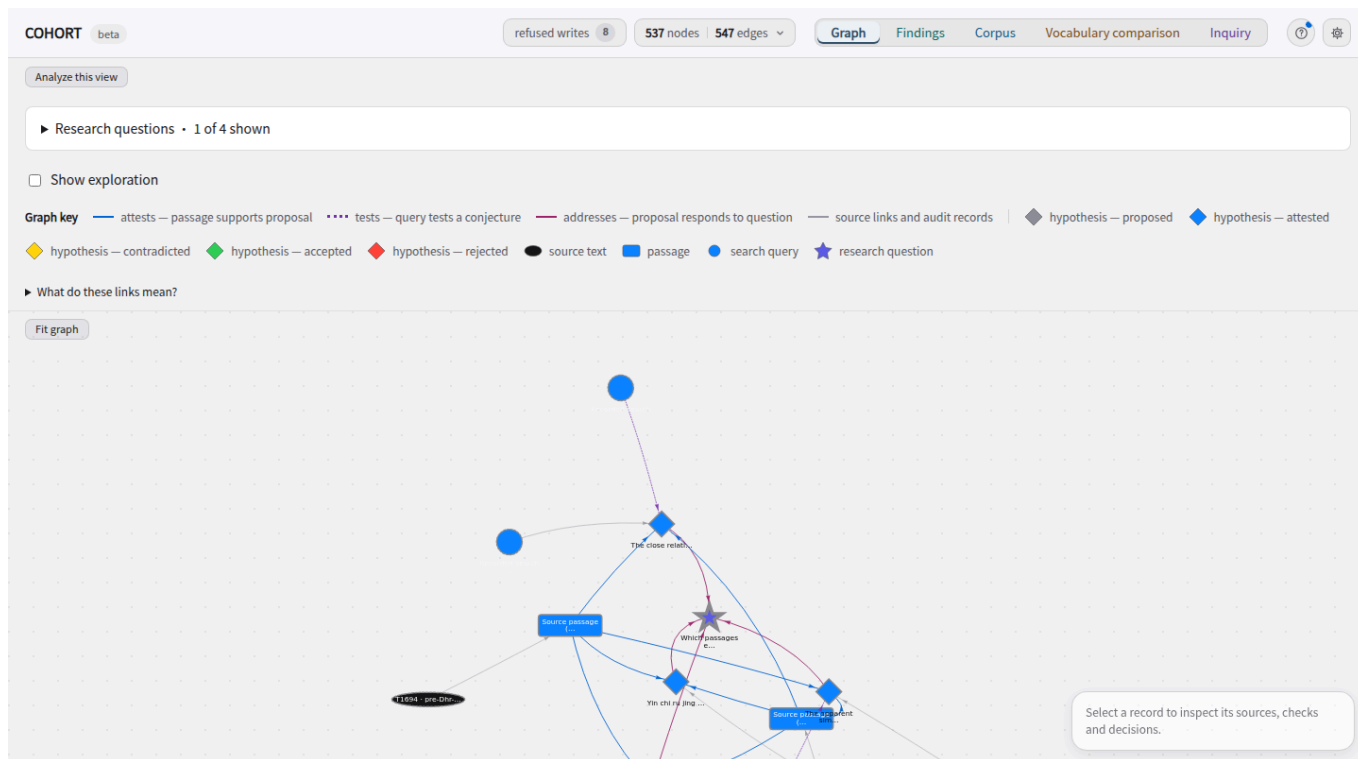
Checks

Read its explanation
and alternatives.

Inspect one proposal at a time.

Follow links between sources, proposals and decisions.

Interface



Show exploration adds recorded tool activity.

Screenshot: source passages hidden where applicable. Diagram: schematic.

How to use it

Select a research question



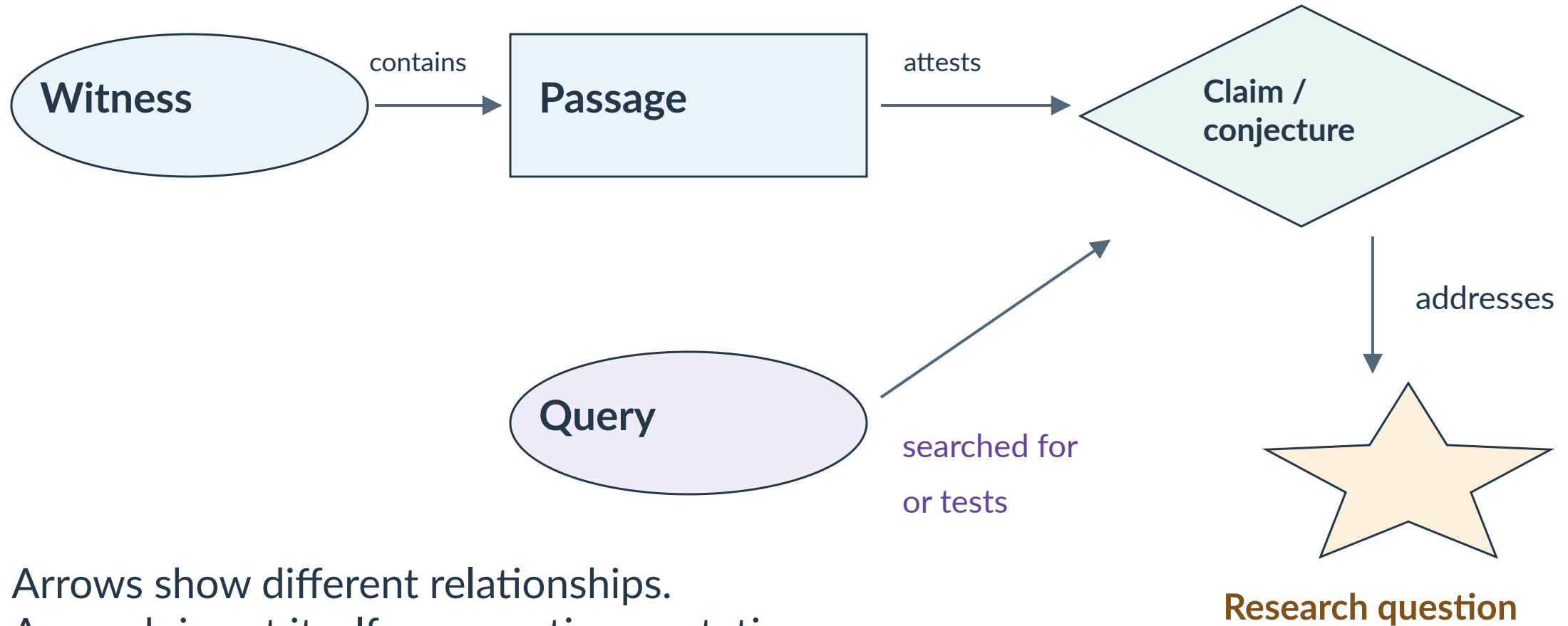
Inspect linked records



Accept or reject a record

Filtering the view does not delete records.

Why a query points to a proposal



Arrows show different relationships.
A search is not itself a supporting quotation.

Why use an evidence graph?

Trace a proposal Follow its citations to passages and source records.

Inspect dependencies See recorded quotation, parallel and descent links.

Keep the history Who proposed it, what was checked, what the researcher decided.

Future work: edge prediction

Suggest a quotation or parallel to investigate, with passages to check.

Cohort

- 1 Find** Lotus passages and shared wording
- 2 Compare** T0603 with and without its commentary
- 3 Inspect** Saved actions, proposal, citations and checks

Results and limits

Useful outputs

Lotus: a related passage with little exact shared wording.

T0603: exclusion exposes the commentary's effect on the ranking.

Proposals, sources and checks can be inspected together.

Observed limits

Kumārajīva's Lotus wrongly ranks Dharmakṣema first in vocabulary.

One known-answer example cannot validate attribution.

Weak tests and reviewer errors still require researcher scrutiny.